microscopic realm, but if ħ happened to be much larger than it is, the strange happenings at the H-Bar would actually be commonplace. As we shall see, their microscopic counterparts certainly are.

What Are the Lumps?

Planck had no justification for his pivotal introduction of lumpy energy. Beyond the fact that it worked, neither he nor anyone else could give a compelling reason for why it should be true. As the physicist George Gamow once said, it was as if nature allowed one to drink a whole pint of beer or no beer at all, but nothing in between.5 In 1905, Einstein found an explanation and for this insight he was awarded the 1921 Nobel Prize in physics.

Einstein came up with his explanation by puzzling over something known as the photoelectric effect. The German physicist Heinrich Hertz in 1887 was the first to find that when electromagnetic radiation —light—shines on certain metals, they emit electrons. By itself this is not particularly remarkable. Metals have the property that some of their electrons are only loosely bound within atoms (which is why they are such good conductors of electricity). When light strikes the metallic surface it relinquishes its energy, much as it does when it strikes the surface of your skin, causing you to feel warmer. This transferred energy can agitate electrons in the metal, and some of the loosely bound ones can be knocked clear off the surface.

But the strange features of the photoelectric effect become apparent when one studies more detailed properties of the ejected electrons. At first sight you would think that as the intensity of the light—its brightness—is increased, the speed of the ejected electrons will also increase, since the impinging electromagnetic wave has more energy. But this does not happen. Rather, the number of ejected electrons increases, but their speed stays fixed. On the other hand, it has been experimentally observed that the speed of the ejected electrons does increase if the frequency of the impinging light is increased, and, equivalently, their speed decreases if the frequency of the light is decreased. (For electromagnetic waves in the visible part of the spectrum, an increase in frequency corresponds to a change in color from red to orange to yellow to green to blue to indigo and finally to violet. Frequencies higher than that of violet are not visible and correspond to ultraviolet and, subsequently, X rays; frequencies